Label-Free Recovery of a Latent Routing Partition from m-Response Profiles

An explanatory walkthrough of the formalization and theory

Working note (expanded version)

June 10, 2026

How to read this document

This is the same formalization as the terse working note, rewritten for understanding. Every formal object is introduced three times: first in plain language (what it is and why we need it), then formally, then instantiated in a *running example* — the Seattle housing scaffold — with concrete numbers. Each theorem is preceded by a "what this says" paragraph and followed by a "why it matters" paragraph. If you read only the plain-language paragraphs and the running example, you will still get the full story; the formal statements are there so that, when we write the paper, we know exactly what has to be proven.

A guide to the structure:

- §1 states the problem with no math at all.
- §2 is a notation table to refer back to.
- §3–4 build the generative model (T1): what the auditor observes and why it has admixture structure.
- §6 works through the two instantiations (the adversarial scaffold and the benign mixture-of-experts).
- §8 says exactly what the method outputs (your "classifier-like" read-outs).
- §9–11 are the four theorem groups (T2, T2′, T3, T4), each explained before and after.
- §13 is an honest table of what is proven, what is sketched, and what is so far only simulation evidence.

1 The problem in plain language

Imagine a company deploys a housing-price model f. You can send it inputs and read outputs — nothing else. You suspect the model does not treat all inputs the same way: internally, it sorts inputs into a small number of *behavioral modes* and behaves differently in each. Two motivating cases:

1. **Adversarial (the Slack scaffold).** The company hides a biased model behind a detector. Inputs that look like real houses get the biased model; inputs that look like an auditor's synthetic probes get an innocuous model. The "mode" of an input is which branch it is routed to, and the routing is designed to make audits come back clean.

2. **Benign (a mixture-of-experts).** A modern model may route each input to one of K expert subnetworks. Nothing is being hidden, but the routing is still invisible from outside, and recovering it tells you how the model organizes its behavior.

In both cases there is a hidden function g (the *gate*) that assigns every input a mode, and the question is:

> *Using only queries to f — no labels, no internals, no trusted reference data — can we recover, for each input we care about, which mode the model put it in?*

The tool is simple: for each input, ask the model not just about the input itself, but also about a handful of slightly perturbed copies of it. The list of answers is the input's *response profile*. The whole theory is about what that profile reveals, and the punchline is that it reveals a great deal, for a structural reason: **each perturbed copy is routed by the gate independently**, so the profile of an input near a gate boundary is a *blend* of two modes' behaviors, and the blend proportions trace out exactly where the input sits relative to the hidden boundary.

2 Notation at a glance

$f : \mathcal{X} \to \mathcal{Y}$	the deployed black box; we can only query it
$g : \mathcal{X} \to [K]$	the hidden gate; $g(x)$ is the mode input x is routed to
K	number of behavioral modes (2 for the scaffold; # experts for an MoE)
$z_i = g(x_i)$	the mode of the i-th queried input — **the thing we want to recover**
f_k	the function the model computes when in mode k
δ_{ij}	the j-th perturbation applied to input x_i
Q_σ	the probe: the distribution perturbations are drawn from, with scale σ
Φ_i	the response profile of x_i: the $(m{+}1)$ answers $(f(x_i), f(x_i{+}\delta_{i1}), \dots)$
π_i	the local routing composition of x_i (Definition 1); a probability vector over modes
Δ^{K-1}	the simplex: all probability vectors over K modes
e_k	the k-th vertex of the simplex (all mass on mode k)
W, H, M	the matrix form: modes, compositions, observed summaries (§5)
$\sigma_{\min}(W),\ \kappa(W)$	smallest singular value and condition number of W; "how distinguishable and well-spread the modes are"
ε	fraction of corrupted (impure) anchors in T4

3 Setup: what the world is and what we observe

The model under audit. $f : \mathcal{X} \to \mathcal{Y}$ is queried as a black box. We *posit* (this is our model of the world, not an assumption we can check directly) that f is organized as a gate plus mode

functions:

$$f(x) \;=\; f_{g(x)}(x),$$

that is: the gate looks at x, picks a mode $g(x) \in \{1, \ldots, K\}$, and the model then answers with that mode's function $f_{g(x)}$ evaluated at x. Observations carry a little noise:

$$\text{observed answer at } x \;=\; f_{g(x)}(x) + \eta, \qquad \mathbb{E}\eta = 0,\ \mathbb{E}\,\|\eta\|^2 = \tau^2_{\text{obs}}. \tag{1}$$

This is the *point-mass* model of mode behavior: given the input, each mode's answer is a fixed number (or vector), not a random draw. It exactly matches both instantiations below — a scaffold's branches and an MoE's experts are deterministic functions. (What changes if modes are genuinely random is deferred to Remark 2; short answer: nothing essential, provided the mode *means* stay separated.)

Running example (introduced). *Seattle housing scaffold.* Inputs x are houses; $b(x) \in [0, 1]$ is the neighborhood's Black/Latino population share; $\tau = 0.4$ a threshold. The model is

$$f(x) = \begin{cases} h(x), & b(x) < 0.4 \quad \text{(mode 1, full valuation)}, \\ h(x) - 0.3, & b(x) \geq 0.4 \quad \text{(mode 2, penalized log-price)}, \end{cases}$$

where $h(x)$ is an honest valuation (in log-price). So $K = 2$, the gate is $g(x) = 1$ if $b(x) < 0.4$ else 2, and the two mode functions are $f_1 = h$ and $f_2 = h - 0.3$. The gate and the 0.3 penalty are hidden; all we can do is query. (Our codebase's defaults are $\tau = 0.25$ with a subtractive 0.30 log-price penalty; $\tau = 0.4$ is used throughout this note purely for illustration.)

The probe and the profile. The auditor picks a *probe*: a recipe for generating small perturbations. The default is Gaussian: $\delta \sim \mathcal{N}(0, \sigma^2 I)$, where σ (the *probe scale*) is the auditor's main knob. For each input x_i we draw m perturbations and query the model $m{+}1$ times:

$$\Phi_i \;=\; \big(\; \underbrace{f(x_i)}_{\text{the } \textit{anchor}} \;,\; f(x_i + \delta_{i1}),\; \ldots,\; f(x_i + \delta_{im}) \,\big). \tag{2}$$

The first coordinate — the model's answer at the *unperturbed* input — is called the **anchor**, and it will play a special role throughout.

Running example (continued). Take a house with $b(x_i) = 0.38$ (just below the threshold) and honest value $h(x_i) = 13.0$. With $m = 20$ and $\sigma = 0.05$ on the b-coordinate, the profile might look like

$$\Phi_i = (\underbrace{13.0}_{\text{anchor}},\ 13.01,\ 12.71,\ 12.99,\ 12.70,\ 13.02,\ \ldots)$$

Notice the two "levels": perturbed copies whose b stayed below 0.4 answer near 13.0 (mode 1), while copies pushed above 0.4 answer near 12.7 (mode 2: the -0.3 penalty). *The hidden gate is visible in the profile as a bimodal pattern.* That observation, made rigorous, is the entire method.

The estimand. Given only the profiles $\{\Phi_i\}_{i=1}^n$, recover the routing partition $z_i = g(x_i)$ for the queried inputs. Two unavoidable caveats, stated now so the theorems are honest later: recovery can only ever be (i) *up to relabeling* of the modes (nothing in the data says which mode is "number 1"), and (ii) *up to behavioral equivalence*: if two modes answer identically on everything the probe can reach, no black-box method can tell them apart.

4 T1: the profile is an induced admixture

The key fact, in words. When we query f at a perturbed point $x_i + \delta$, the gate does not know or care that this query is "about" x_i. It simply routes the point it is given: the answer is governed by $g(x_i + \delta)$, **not** by $g(x_i)$. If x_i sits near a gate boundary, some perturbations land on the other side and get the other mode's answer. So a single profile mixes modes — and the mixing proportions encode how close x_i is to the boundary. We give that proportion a name.

Definition 1 (Local routing composition)**.** The local routing composition of x_i under probe Q is the probability vector $\pi_i \in \Delta^{K-1}$ with entries

$$\pi_i(k) \;:=\; \mathbb{P}_{\delta \sim Q(\cdot|x_i)}\big[g(x_i + \delta) = k\big], \qquad k = 1, \ldots, K. \tag{3}$$

In words: $\pi_i(k)$ is the fraction of x_i's probe-neighborhood that the gate assigns to mode k.

Running example (the three houses). With threshold $\tau = 0.4$ and probe scale $\sigma = 0.05$ on the b coordinate:

- **House A**, $b = 0.10$ (deep in mode 1): essentially no perturbation reaches 0.4, so $\pi_A \approx (1,\, 0)$. Its profile is flat: all $m{+}1$ answers near $h(x_A)$.
- **House B**, $b = 0.38$ (near the boundary, mode-1 side): the fraction of perturbations crossing is $\mathbb{P}[\,0.38 + \delta \geq 0.4\,] = \Phi\big(\frac{0.38-0.4}{0.05}\big) \approx 0.34$, so $\pi_B \approx (0.66,\, 0.34)$. Its profile is the bimodal one shown above.
- **House C**, $b = 0.42$ (just past the boundary, mode-2 side): $\pi_C \approx (0.34,\, 0.66)$, the mirror image; its *anchor* sits at the penalized level $h(x_C) - 0.3$.

So the composition π_i is a smooth function of where the house sits: it slides from $(1, 0)$ to $(0, 1)$ as b crosses the threshold, with the transition width set by σ. **This is what we mean by the compositions being "geometry-coupled": they are not random — they are a deterministic function of position relative to the hidden boundary.**

The generative model. Putting it together: coordinate j of the profile is generated by first letting the gate route the perturbed point ($a_{ij} = g(x_i + \delta_{ij})$, which is a draw from Categorical(π_i) over the probe's randomness), and then answering with that mode's function.

Definition 2 (Induced separable admixture)**.** The profile is distributed as

$$\Phi_i \;\sim\; \underbrace{\delta_{f_{z_i}(x_i)}}_{\textit{pure anchor}} \;\otimes\; \bigotimes_{j=1}^{m} \underbrace{\Big[\textstyle\sum_{k=1}^{K} \pi_i(k)\, \delta_{f_k(x_i + \delta_{ij})}\Big]}_{\text{mixture with weights } \pi_i} + \text{noise}. \tag{4}$$

Reading the symbols: δ_v is the point mass at value v ("the answer is exactly v"), and $\otimes$ just means the coordinates are generated independently. The first factor says the anchor is *exactly* mode z_i's answer — the unperturbed input cannot have crossed any boundary, so the anchor is a **pure** observation of the input's own mode. Each remaining coordinate is a *mixture* across modes with weights π_i.

Why this is called an "admixture" and why it matters. In topic-model language: each input is a "document," the modes are "topics," and the profile coordinates are "words" drawn from a document-specific blend π_i of the topics. Models of this shape are called admixtures. Two features make ours special, and both help us:

1. **The blend weights are induced, not assumed.** In topic models the proportions are postulated as random (usually i.i.d. Dirichlet). Here they are *forced by the query mechanism* and are a deterministic function of the gate geometry and the probe (Definition 1). This will matter when we choose which recovery theorem to use: we need one that makes *no* distributional assumption on the weights (§9).

2. **There is a guaranteed-pure coordinate.** Topic models must *assume* some purity (the famous "anchor word" assumption). We get a pure coordinate for free, per input: the anchor. Purity is what makes the factorization identifiable.

5 The matrix form: turning profiles into a factorization

In words. To analyze recovery we summarize each profile as a vector and stack the vectors into a matrix. The summary is taken through a *feature map* ψ — think of ψ as binning the answers into V buckets, so a profile becomes a histogram over answer-levels. In the running example with two levels (h and $h - 0.3$ after anchor-differencing), V can be as small as 2 or 3.

Each mode k then has a *signature vector* $w_k \in \mathbb{R}^V$: the expected histogram of answers produced by mode k. Stack the signatures as columns of $W \in \mathbb{R}^{V \times K}$ (the **mode matrix**), and the compositions as columns of $H \in \mathbb{R}^{K \times n}$ (the **composition matrix**). Summarizing input i's m perturbed answers by their empirical histogram $M_{\cdot i}$ gives

$$M \;=\; W\,H \;+\; N, \qquad M \in \mathbb{R}^{V \times n}, \tag{5}$$

where the error N shrinks as m grows ($\|N_{\cdot i}\| \approx \sqrt{(\tau_{\text{obs}}^2 + \sigma_w^2)/m}$, with σ_w^2 the within-mode answer variability over the small probe neighborhood).

What recovery means geometrically. Equation (5) says: every observed column $M_{\cdot i}$ is (approximately) a convex combination $W\pi_i$ of the K mode signatures. So the data columns live inside a $(K-1)$-dimensional *simplex* whose corners are the mode signatures $w_1, \ldots, w_K$. For $K = 2$: all columns lie on a *segment* whose endpoints are the two modes; an input's position along the segment *is* its composition. Recovery is therefore a geometry problem:

> **find the corners of the simplex the data lives in; then read off each input's blend coordinates relative to those corners.**

This is exactly the problem nonnegative matrix factorization (NMF) solves, and "the corners are present in the data" is what NMF calls *separability*.

6 The two instantiations

6.1 The adversarial gate: feature-thresholded redlining ($K = 2$)

Our implemented scaffold gates not on manifold membership but on something blunter: the protected feature itself,

$$f(x) \;=\; h(x) \;-\; 0.3 \cdot \mathbf{1}\big[\, b(x) \geq \tau \,\big], \qquad g(x) = \mathbf{1}[\, b(x) \geq \tau \,] \in \{1, 2\},$$

with $f_1 = h$ (full valuation) and $f_2 = h - 0.3$ (penalized) — exactly the running example. Structural facts:

- **Anchor purity is automatic.** Any gate that is a deterministic function of *location* routes the unperturbed input by its own position, so the anchor is pure with no help from the attacker's design. (ε in T4 then models measurement noise, gate randomness, or future adaptive behavior — not a property we depend on the adversary to grant.)
- **The estimand is the redlining map.** Real houses sit on both sides of $b = \tau$, so the partition over real data is genuinely non-trivial: recovering z_i *is* recovering which houses receive the hidden penalty — the per-instance object the auditing literature does not provide.
- **"Adversarial" is intent, not geometry.** Structurally this gate is identical to the benign competence switch of §6.3 — a threshold on a scalar function of the input. The recovery mathematics sees only the geometry; what makes the case adversarial is concealment and harm. This is a feature of the framework: one theory covers both instantiations, and the difference between them lives on the alignment axis of §7, not in the theorems.

Relation to the original Slack scaffold. Slack's gate is a *probe-detection* gate: an OOD detector $d(x)$ routes "real" inputs to the biased model and suspected synthetic queries to an innocuous one. Under that gate every real input is mode-biased, so the per-instance partition over real data is trivial; the informative target there is which *queries* get routed. The full fooling-LIME/SHAP construction *stacks* the two gates, giving $K = 3$ effective modes: (real, $b < \tau$) $\to h$; (real, $b \geq \tau$) $\to h - 0.3$; (flagged probe) $\to$ innocuous. The formalism covers the stack as a $K = 3$ gate; the two boundaries are crossed by *different components of the probe*, which is precisely what the probe-relative alignment of §7 captures.

6.2 The benign mixture-of-experts (general K): signal lives at boundaries

An MoE routes via $g(x) = \arg\max_k r_k(x)$ to one of K experts f_k. The anchor is again pure (it is the answer of whichever expert owns x_i). But the geometry differs in one important way:

> **Deep inside an expert's region there is nothing to mix.** If every perturbation of x_i stays inside expert z_i's region, then $\pi_i = e_{z_i}$ exactly — the composition sits at a simplex corner and the profile is unimodal. Such an input is trivially *consistent* with its expert but its profile alone cannot distinguish "expert 3" from "expert 5" in any absolute sense; only inputs near *router boundaries* produce mixed profiles that tie the experts' signatures together.

So benign recovery has a two-zone character: boundary-adjacent inputs are recovered *directly* (their mixed profiles reveal the local pair of modes), and interior inputs are recovered *by propagation* — they share a signature level with some boundary-adjacent input of the same expert. This asymmetry is not a flaw; it is the honest statement of where the information is, and it is also why the deep-instance corners are exactly what makes the factorization separable (the corners of the simplex are *populated*). Finally, in the MoE case K is unknown and must be inferred (§12).

6.3 A third benign example: a competence switch

A served system routes easy inputs to a small fast model and hard ones to a large model: $g(x) = \mathbf{1}[c(x) < \theta]$ for a confidence score c. Everything above applies with $K = 2$; the difference from the

scaffold is purely *geometric* — where the boundary sits relative to the data — which is the subject of the next section.

7 One axis to unify both cases: gate–manifold alignment

In words. Real data does not fill input space; it concentrates near a lower-dimensional set $\mathcal{M}$ (the *data manifold*: the set of "realistic" inputs). A gate boundary can relate to that set in two extreme ways:

- It can cut *across* the data — like our feature-threshold gate: the level set $\{b(x) = \tau\}$ of a real, varying feature. The boundary's normal points along the b-axis, a direction in which realistic inputs genuinely differ, so small realistic perturbations cross it easily.
- It can run *along* the data — like an MoE competence boundary that follows level sets of difficulty. Small realistic perturbations slide along it and rarely cross.
- It can *wrap around* the data — like Slack's OOD-detector boundary, which encloses the realistic region. Its normal points *off-manifold*: realistic (on-manifold) perturbations never cross it at all, yet a full-ambient Gaussian probe crosses it readily, because in high ambient dimension most of the probe's mass moves off-manifold. This case shows that alignment with the *manifold* is not the whole story; what ultimately matters is alignment with the *probe*.

Definition 3 (Gate–manifold alignment)**.** At a boundary point x with boundary normal $\nu(x)$ (the direction perpendicular to the boundary), define

$$\alpha(x) \;=\; \left\| \operatorname{proj}_{T_x\mathcal{M}} \nu(x) \right\| \;\in [0,1],$$

the length of the normal's projection onto the data manifold's tangent space $T_x\mathcal{M}$ (the directions a realistic perturbation can move). $\alpha \approx 1$: the crossing direction is available to realistic probes (scaffold-like). $\alpha \approx 0$: it is not (competence-like).

Why we bother, and the general form. α is the dial that interpolates the two instantiations: "one method, two case studies" becomes "one method, one axis." The fully general quantity, prompted by the wrap-around case above, is the *probe-relative crossing scale*: for a probe with covariance Σ and boundary normal $\nu(x)$,

$$\sigma_{\text{eff}}(x) \;=\; \sqrt{\nu(x)^\top \Sigma\, \nu(x)},$$

the probe's standard deviation *in the crossing direction*. The crossing probability at signed distance s is then $\approx \Phi(-s/\sigma_{\text{eff}})$. Manifold alignment is the special case of a probe supported on realistic directions, where $\sigma_{\text{eff}} = \sigma\,\alpha$. Two consequences: all recoverability statements below depend on gate and probe only through σ_{eff}; and the probe's *support*, not just its scale, is an identifiability knob — an on-manifold probe can never recover a wrap-around gate ($\sigma_{\text{eff}} = 0$) that an ambient probe recovers easily. This is a second, qualitative instance of T3's thesis that probe design changes what is identifiable.

8 What the method outputs

The estimator's native output, per input, is a *soft composition* $\widehat{\pi}_i \in \Delta^{K-1}$ — "this input's neighborhood is 66% mode 1, 34% mode 2." Everything classifier-like is a read-out of it:

- **Regime classification:** $\widehat{z}_i = \arg\max_k \widehat{\pi}_i(k)$. This is the hard partition — "which regime is this input in" — recovered up to a global relabeling of modes.
- **Flagging bad points:** pick the adverse mode $k^\star$ (in the scaffold: the biased branch) and flag input i iff $\widehat{\pi}_i(k^\star) \geq t$. This is the per-instance "this point is treated adversarially" alarm.
- **Boundary proximity for free:** how far $\widehat{\pi}_i$ sits from a corner measures how close the input is to a gate boundary — useful diagnostic information that a hard classifier would throw away.

We state all guarantees for $\widehat{\pi}_i$; the two classifier read-outs inherit them automatically (Corollary 1).

9 T2 and T2′: identifiability and a working estimator

What T2 says, in words. *If the modes genuinely differ, and the inputs' compositions are sufficiently spread out over the simplex, then the profiles determine the compositions and the mode signatures uniquely (up to relabeling).* In other words: the recovery problem is well-posed — the answer is actually contained in the data we collect. This is the theorem that decides whether the project makes sense at all.

We need three assumptions, each with a plain meaning:

Assumption 1 (Recovery conditions)**.**

(i) **Separated modes.** The mode matrix W has full column rank, with smallest singular value $\sigma_{\min}(W) > 0$. *Meaning: the K modes have genuinely different answer signatures; no mode's behavior is a blend of the others'. In the running example: $f_1 - f_2 \equiv 0.3 \neq 0$.*

(ii) **Composition diversity.** The matrix $G = \frac{1}{n} HH^\top$ is nonsingular; write $\kappa(H)$ for H's condition number. *Meaning: the inputs' blend vectors are spread out — not all sitting at one corner, not all collapsed at the centroid. This is the assumption the probe controls (T3).*

(iii) **Anchor purity.** Anchors are pure draws of their own mode. *(Relaxed to a $1 - \varepsilon$ fraction in T4.)*

Theorem 1 (T2: exact identifiability, noiseless)**.** *Under Assumption 1 with no noise ($N = 0$), the factorization $M = WH$ determines W and H uniquely up to one global permutation of the mode labels.*

Proof, with the geometry spelled out. Every data column $M_{\cdot i} = W\pi_i$ is a convex combination of the columns of W, so the data lives in the simplex $\mathrm{conv}\{w_1, \ldots, w_K\}$. Separability — supplied by the pure anchors and the deep instances, which sit *at* the corners — means each corner is itself realized by some data column. A nondegenerate simplex (full column rank of W, Assumption (i)) is determined by its corner set, and corners are exactly the points of the set that are not convex combinations of the others; hence the data determine the corner set, i.e. W, up to the order in which we list the corners. Given the corners, each input's blend coordinates π_i are the unique barycentric coordinates of $M_{\cdot i}$ in the simplex (uniqueness again by full rank). The only ambiguity left is the listing order of corners — the global label permutation. □

What T2′ adds, in words. T2 is about information; T2′ is about computation and noise. It says a *specific, fast algorithm* — the Successive Projection Algorithm (SPA), which greedily picks the most extreme data columns as corner estimates — recovers the corners even when the data is noisy, with an error bound, and crucially **without any probabilistic assumption on the compositions**. That last clause is the entire reason we route the theory through separable NMF rather than through topic-model (Dirichlet) machinery: our compositions are deterministic and geometry-coupled, which would violate a Dirichlet assumption but is perfectly admissible here.

Theorem 2 (T2′: robust recovery via SPA)**.** *Let $\widehat{W}$ be the corners selected by SPA on the columns of $M = WH + N$, and let $\widehat{\pi}_i$ solve the simplex-constrained least-squares fit of $M_{\cdot i}$ against $\widehat{W}$. If Assumption 1 holds and the per-column noise obeys*

$$\max_i \|N_{\cdot i}\| \;\le\; \epsilon \;\le\; \mathcal{O}\!\left(\frac{\sigma_{\min}(W)}{\sqrt{K}\,\kappa^2(W)}\right),$$

then every corner is recovered to accuracy $\mathcal{O}(\epsilon\,\kappa^2(W))$ and

$$\max_i \;\|\widehat{\pi}_i - \pi_i\| \;\le\; \mathcal{O}\!\left(\frac{\epsilon\,\kappa^2(W)}{\sigma_{\min}(W)}\right).$$

Since $\epsilon = \mathcal{O}\big(\sqrt{(\tau_{\text{obs}}^2 + \sigma_w^2)/m}\big)$, the error vanishes as the number of perturbations m grows.

Proof sketch, with the logic of each step. Step 1 (cite). The corner-recovery claim is the Gillis–Vavasis robustness theorem for SPA on near-separable matrices. Its hypotheses are: W full column rank; corners present among the columns (separability); column noise below the stated threshold. Note what is *absent*: any condition on how the columns of H are distributed. Deterministic, correlated, geometry-coupled compositions satisfy the hypotheses verbatim. *Step 2 (propagate).* With corners known to accuracy $\mathcal{O}(\epsilon\kappa^2(W))$, the barycentric-coordinate solve is a least-squares problem whose sensitivity is $1/\sigma_{\min}(W)$; multiplying gives the $\widehat{\pi}_i$ bound. *Step 3 (noise scale).* Each column of M is an average of m bounded summaries, so its deviation concentrates at rate $1/\sqrt{m}$ — this is ordinary Monte-Carlo averaging. □

Corollary 1 (The classifier read-outs inherit the guarantee)**.** *If $\max_i \|\widehat{\pi}_i - \pi_i\| \le \gamma$, then (a) every input whose true top mode has margin $\pi_i(z_i) \ge \frac{1}{2} + \gamma$ is classified correctly by $\widehat{z}_i = \arg\max_k \widehat{\pi}_i(k)$ (up to the global permutation); and (b) the flag at threshold t makes no mistakes on inputs whose true $\pi_i(k^\star)$ is at least γ away from t.*

Why this pair is the hinge of the project. T2 says the problem is worth attempting (the answer is in the data); T2′ says a real polynomial-time estimator gets it, with explicit constants saying *when*. Everything empirical now has a home: the simulations check whether realistic gates satisfy Assumption 1(ii) — and that is precisely what the probe controls, which brings us to T3.

10 T3: probe design changes what is identifiable

The claim, in words. The conditioning of H — Assumption (ii), the thing recovery quality divides by — is not a fact of nature. It is *manufactured by the auditor's probe*. Turn the probe-scale dial σ and you move the compositions around the simplex:

- σ too small: no perturbation ever crosses the boundary; every π_i sits exactly at its corner e_{z_i}. The profiles are unimodal, the blend information is gone. (House B at $\sigma = 0.001$: $\pi_B = (1, 0)$, indistinguishable in shape from deep House A.)

- σ too large: every neighborhood spans all modes about equally; every π_i collapses toward the same centroid. All inputs look alike; the matrix H approaches rank one and $\kappa(H) \to \infty$.

- σ in a middle band: the compositions spread out over the simplex — corners populated by deep inputs, interior populated by boundary-adjacent ones — and H is well-conditioned.

Proposition 1 (Conditioning is probe-controlled)**.** *For a gate with alignment α and Gaussian probe Q_σ, the smallest eigenvalue of $G = \frac{1}{n} H H^\top$ is bounded away from 0 for σ in a band $[\sigma_{\min}(\alpha), \sigma_{\max}(\alpha)]$ and tends to 0 at both ends; the band widens as $\alpha \to 1$ and narrows (eventually to nothing) as $\alpha \to 0$.*

Proof sketch. For an input at signed distance s_i from the boundary, the crossing fraction is, to first order, $\pi_i(\text{other side}) = \Phi\big(-s_i/(\sigma\,\alpha_{\text{eff}})\big)$ with α_{eff} monotone in α: a sigmoid in s_i whose width is $\sigma\alpha_{\text{eff}}$. The spread of the compositions over the simplex is the spread of this sigmoid evaluated on the empirical distances $\{s_i\}$: a too-narrow sigmoid (small $\sigma\alpha$) saturates — everything at the corners; a too-wide one (large $\sigma\alpha$) flattens — everything at the centroid. Matching the sigmoid width to the spread of $\{s_i\}$ gives the band, and the α-dependence follows because α multiplies σ everywhere. □

Corollary 2 (Identifiability is probe-relative)**.** *There are gates and input sets that are* unrecoverable *under a badly-scaled probe yet* recoverable *under a well-scaled probe of the same family. Probe design therefore changes what is identifiable, not merely the error bars at fixed identifiability.*

Why this is the paper's distinctive claim. Every neighboring literature (topic models, slice discovery, fairness audits) takes its "views" of the data as given. Here the auditor *manufactures* the views and can tune them — and the theorem says the tuning decides identifiability itself. This claim has no analogue in the neighbors, and it is also empirically grounded: in simulation, the eigenvalue spectrum of G degrades from condition number ≈ 1 to $\approx 4{,}000$ exactly as σ leaves the band.

11 T4: what happens when anchors are corrupted

The setting, in words. Everything above trusts the anchor. The natural attack — and the bridge to the adaptive-adversary sequel — is to corrupt it: suppose a fraction ε of anchors are *impure* (answered by the wrong mode). What breaks, and how fast?

Assumption 2 (ε-impure anchors)**.** Each anchor is independently pure with probability $1 - \varepsilon$ and answered by a wrong mode with probability ε.

Proposition 2 (Degradation, inversion, and the floor)**.** *Under Assumptions 1(i,ii) and 2:*

1. ***Graceful regime.*** *For ε below a threshold $\varepsilon^\star$ set by the corner separation, the corner estimates move by only $\mathcal{O}(\varepsilon)$ and the composition error grows by an additive $\mathcal{O}(\varepsilon)$.*

2. ***Permutation survival.*** *For all $\varepsilon \neq \frac{1}{2}$ the recovered* partition *(who is grouped with whom) remains consistent; what degrades is the* orientation *(which group gets which name). As $\varepsilon \to 1$ the labels invert outright, and permutation-corrected accuracy returns toward 1.*

3. ***Identifiability floor.*** *At exactly $\varepsilon = \frac{1}{2}$ the anchor carries zero orientation information, and mode* labels *are unidentifiable from anchors — though the unlabeled grouping can survive on the geometry of H alone.*

The intuition for the strange middle claim. The anchor functions as a noisy vote about its input's identity. Votes with flip-rate $\varepsilon < \frac{1}{2}$ are right on average; votes with $\varepsilon > \frac{1}{2}$ are *reliably wrong*, which is just as informative once you know to negate them — the recovered grouping is intact, with names swapped. Only at exactly $\frac{1}{2}$ is the vote pure noise. In simulation this shows up vividly: labeled accuracy collapses through $\varepsilon \approx 0.2$ while permutation-corrected accuracy climbs back to ≈ 0.98 by $\varepsilon = 0.4$, with a sharp dip to chance at 0.5. **Since the estimand was always "the partition up to relabeling," the right reading is: the partition is far more robust than the labels.**

Remark 1 (Bridge to the adaptive adversary). Here ε is static and random. The follow-up paper lets an adversary *choose* which anchors to corrupt, subject to a utility constraint (the attack must still deliver its biased behavior to its targets). The empirically effective defense — true boundary crossings are one-sided in the *direction* of δ, while corrupted anchors produce direction-symmetric discrepancies — relies on the auditor's own probe randomness, which a static gate cannot condition on. Formalizing that probe-vs-gate game is deliberately out of scope here.

12 Inferring the number of modes

In the scaffold, $K = 2$ is known. In the MoE, K is the number of experts and unknown. The geometric picture supplies the estimator: K is the number of corners of the simplex the data spans. SPA-type algorithms add corners greedily; the stopping rule is to stop when the residual — the distance from the data to the simplex spanned so far — drops below the noise floor ϵ of Theorem 2. Under Assumption 1 this recovers the true K, with the usual eigengap-style caveat: confident selection needs the K-th residual drop to be clearly larger than the $(K{+}1)$-th. We therefore state the main theorems for known K and treat K-inference as this residual-threshold corollary.

13 Honest status of each claim

Claim	Content	Status
T1	Induced separable admixture; pure anchor; geometry-coupled compositions	Structural; follows from the query mechanism. Solid.
T2	Noiseless identifiability up to permutation	Proven above (simplex geometry); standard.
T2′	SPA recovery with explicit conditioning; no i.i.d. assumption on H	Reduction to Gillis–Vavasis; the coupling of our noise model to their hypothesis is *sketched, not ground out.* Main proof task.
T3	Conditioning of H is set by (σ, α); identifiability is probe-relative	Proven in shape for 1-D, two-mode boundaries; matches simulation (condition number $1 \to 4{,}000$). General K, multi-dim: open.
T4	Graceful degradation; inversion; floor at $\varepsilon = \frac{1}{2}$	Clean for $K = 2$ (binary flip). For $K \geq 3$ "inversion" must be restated (corruption can scramble, not just swap). Simulation evidence strong at $K = 2$.
K-inference	Residual-threshold stopping rule	Standard given T2′; eigengap caveat applies.

Remark 2 (Distributional generalization). If modes are genuinely random rather than point masses, replace each $f_k(x)$ by the mode's *mean* summary; all theorems go through provided the means stay separated ($\sigma_{\min}(W) > 0$). The fundamental ceiling is then exactly *behavioral equivalence*: two modes with identical mean signatures under the chosen feature map are unidentifiable — and by T3, enriching the probe or the feature map can split equivalence classes that a weaker probe cannot.